

MP491 2020/2021: Tutorial Sheet 3

2-dim nonlinear ODEs

I. Linearisation

For each of the following non-linear ODE systems, locate and classify all equilibrium points according to the Linearisation analysis.

1.

$$\begin{aligned}\frac{dx}{dt} &= x(1 + x - 2y) \\ \frac{dy}{dt} &= y(x - 1)\end{aligned}$$

2.

$$\begin{aligned}\frac{dx}{dt} &= 4 - 4x^2 - y^2 \\ \frac{dy}{dt} &= 3xy\end{aligned}$$

II. Isoclines

For each of the following non-linear ODE systems, locate and classify all equilibrium points. Then use the isoclines for the **full non-linear problem** to sketch the phase portrait in each case. It is sufficient to use the $k = 0, \infty, -1$ and 1 isoclines.

1.

$$\begin{aligned}\frac{dx}{dt} &= -x - y \\ \frac{dy}{dt} &= -x + y + 2xy\end{aligned}$$

2.

$$\begin{aligned}\frac{dx}{dt} &= x - y - 2xy \\ \frac{dy}{dt} &= 4x + y\end{aligned}$$

III. Hamiltonian systems

Show that the following non-linear system is Hamiltonian, find the corresponding Hamiltonian function, locate and classify all equilibrium points, and sketch the phase plane curves in the phase plane.

1.

$$\begin{aligned}\frac{dx}{dt} &= y \\ \frac{dy}{dt} &= 2 - x - 3x^2\end{aligned}$$